ABSTRACT

Visually impaired individuals face significant challenges in navigating outdoor environments due to limited awareness of surrounding hazards and warning cues. Conventional mobility aids such as white canes primarily provide tactile feedback for ground-level obstacles and lack the ability to interpret environmental warning signs or elevated hazards. To address these limitations, this project presents the development of a **Smart Wearable Hazard Detection System** that integrates vision-based hazard recognition with intuitive wearable feedback.

The proposed system consists of two primary modules: a goggle-mounted vision module and a wrist-worn feedback and control module. A HuskyLens AI vision sensor is employed to detect predefined road hazards, including warning markers, stop signs, and traffic cones. Upon detection, hazard information is transmitted to an ESP32-based wristband controller, which processes the input and generates appropriate user alerts. Multimodal feedback is delivered through vibration patterns, audible signals, and visual indicators to ensure timely and intuitive hazard awareness.

A hierarchical control approach is adopted, where vision-based perception is treated as an intelligent sensing layer and alert management is handled by a centralized wearable controller. This project demonstrates effective sensor integration, task partitioning, and wearable system design, fulfilling the requirements of an advanced Mechatronics System Integration (MSI) application aimed at enhancing safety and environmental awareness for visually impaired users.

1. INTRODUCTION

Visually impaired individuals encounter continuous challenges in safely navigating outdoor and urban environments due to the absence of visual feedback regarding surrounding hazards. These challenges include difficulty recognizing warning signs, traffic-related cues, and temporary obstacles commonly found in pedestrian areas. In unfamiliar or crowded environments, delayed hazard awareness can significantly increase the risk of accidents and injuries. According to the World Health Organization, visual impairment affects millions of individuals worldwide, emphasizing the importance of developing assistive technologies that enhance independent mobility and safety.

Traditional mobility aids, such as the white cane, remain widely used due to their simplicity and reliability. However, these aids are primarily limited to detecting obstacles at ground level through physical contact and do not provide information about elevated hazards or contextual environmental cues. Warning signs, traffic cones, and temporary road markers are often positioned above ground level and may not be detected in time using conventional aids alone. As a result, visually impaired users may remain unaware of potential dangers until they are in close proximity to the hazard.

Recent developments in assistive technology have explored the integration of electronic sensors and wearable devices to enhance environmental perception. Vision-based systems, in particular, have shown promise in enabling the recognition of specific objects and symbols within the user's surroundings. When incorporated into wearable platforms, such systems can provide contextual awareness by identifying critical warning cues and delivering timely alerts to the user. Compared to distance-only sensing approaches, vision-based hazard recognition allows for more meaningful interpretation of the environment.

This project proposes a **Smart Wearable Hazard Detection System** designed to assist visually impaired users by detecting and interpreting environmental warning cues using a goggle-mounted vision sensor. The system employs a HuskyLens AI vision module to recognize predefined hazard indicators such as warning markers, stop signs, and traffic cones. Detected hazard information is transmitted to a wrist-worn ESP32-based controller, which generates intuitive feedback through vibration, audio alerts, and visual indicators. By adopting a wearable and modular design, the proposed system aims to enhance hazard awareness while maintaining comfort, portability, and suitability for daily use. The system architecture demonstrates hierarchical control and sensor integration principles, aligning with the core objectives of an advanced Mechatronics System Integration (MSI) project.

Table 1. Comparison of Existing Assistive Mobility Systems and the Proposed System

Reference	System Description	Key Features	Limitations	Proposed System Advantage
[1]	Conventional white cane	Ground-level tactile feedback	No detection of elevated hazards or warning cues	Vision-based hazard recognition
[2]	Ultrasonic-based smart cane	Distance-based obstacle alerts	No contextual understanding of environment	Semantic hazard interpretation
[3]	Wearable electronic travel aid	Wearable form factor, audio alerts	Limited object classification capability	AI-based visual recognition

[4]	Vision-assisted navigation system	Camera-based object detection	High computational complexity, bulky setup	Lightweight vision sensor with wearable feedback
Proposed System	Smart wearable hazard detection system	Vision-based hazard detection, wearable feedback, hierarchical control	Prototype-scale implementation	Improved contextual awareness with modular MSI architecture

2. METHODOLOGY

2.1 Project Development

Overall, a block diagram is an overview of the entire system such that the connection between input, output and microcontroller can be explained. The project on the Smart Wearable Hazard Detection System has 2 circuits which are categorized as system 1 and system 2 . The block diagram of the system 1 (Smart Goggles) is displayed in Fig. 1. While Fig. 2 will show the block diagram of System 2 (Smart Bracelet) . The following single diagram demonstrates the collaboration of the two mechanisms that secure the efficiency of the project in general. Fig. 3 indicates the flow chart of the system 1 and system 2 that represents the work principle of the Smart Wearable Hazard Detection System project.

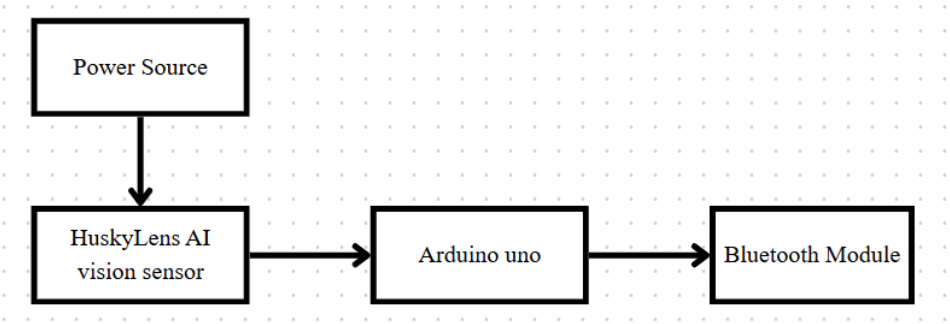

Fig. 1 Block diagram of system 1 Smart Goggles.

Fig. 2 Block diagram of system 2 Smart Bracelet.

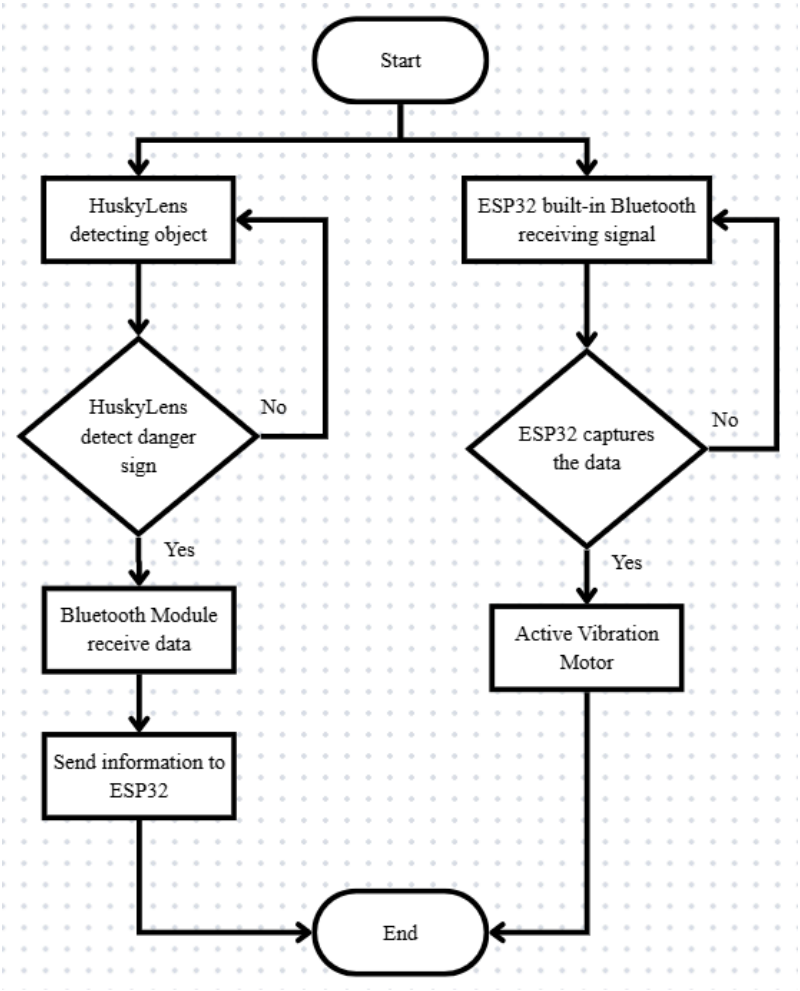

Fig. 3 The flow chart that explains all the work principles for the Smart Wearable Hazard Detection System project.

2.2 Schematic Diagram and Prototype Setup

Fig. 2.2a exhibits the schematic design of the Smart Goggles and Smart Bracelet systems, illustrating the wearable integration of the vision and haptic components. Fig. 2.2b is the circuit diagram of System 1 (Smart Goggles), which utilizes a Huskylens AI camera for object recognition and an Arduino Uno for logic processing. The system is powered by a18650 Li-ion batteries regulated through a TP4056 charging module and an MT3608 Boost Converter to provide a stable 5V supply. Wireless data transmission is handled via an HC-05 Bluetooth module, featuring a voltage divider to protect the module's 3.3V logic pins. Fig. 2.2c exhibits the circuit diagram of System 2 (Smart Bracelet), a haptic feedback receiver designed for wearable alerts. This system utilizes an ESP32 microcontroller with integrated Bluetooth to receive trigger signals from the goggles. It is powered by a 3.7V Li-ion battery configuration managed by a TP4056 module. To provide tangible feedback, the circuit includes a vibration motor driven through a transistor-based switching circuit to ensure the motor's current draw does not exceed the ESP32's GPIO limits.

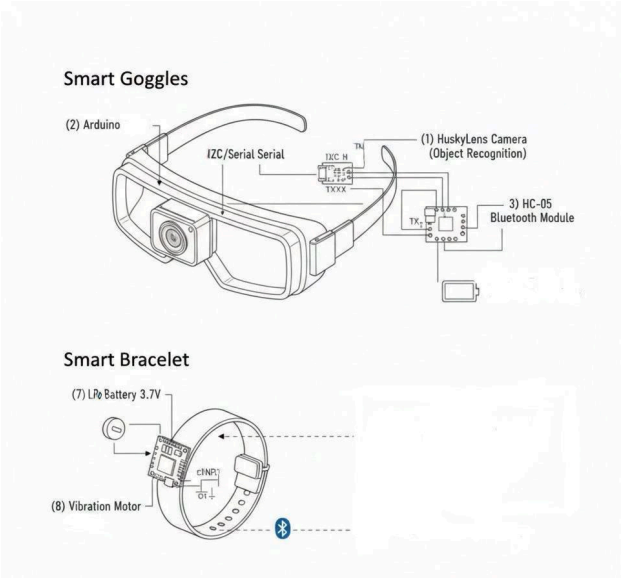

Fig 2.2a

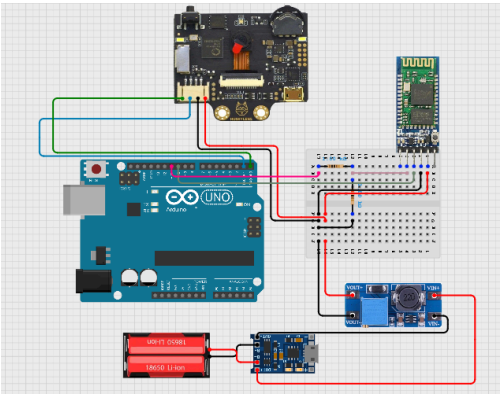

Fig 2.2b

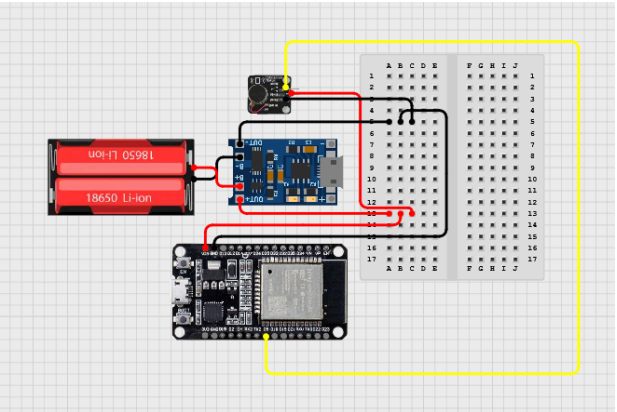

fig 2.2c

REFERENCES

[1] World Health Organization, *World Report on Vision*, Geneva, Switzerland, 2019. Available: https://www.who.int/publications/i/item/world-report-on-vision

[2] A. V. Nandini, A. Dwivedi, N. A. Kumar, T. S. Ashwin, V. Vishnuvardhan, and R. M. R. Guddeti, "Smart Cane for Assisting Visually Impaired People," *IEEE Region 10 Conference (TENCON)*, Kochi, India, 2019, pp. 546–551, doi: 10.1109/TENCON.2019.8929579.

[3] R. Velázquez, "Wearable Assistive Devices for the Blind," *Wearable and Autonomous Biomedical Devices and Systems for Smart Environment*, Springer, Berlin, Heidelberg, 2010, pp. 331–349, doi: 10.1007/978-3-642-12644-3_15.

[4] S. Shoval, J. Borenstein, and Y. Koren, "Auditory Guidance With a Portable Obstacle Detection Device for the Blind," *IEEE Transactions on Systems, Man, and Cybernetics*, vol. 28, no. 3, pp. 459–467, Aug. 1998, doi: 10.1109/3468.686714.

[5] P. Meijer, "An Experimental System for Auditory Image Representations," *IEEE Transactions on Biomedical Engineering*, vol. 39, no. 2, pp. 112–121, Feb. 1992, doi: 10.1109/10.121642.

[6] M. F. Razali, S. F. Toha, and Z. Z. Abidin, "Intelligent Path Guidance Robot for Visually Impaired Assistance," *Procedia Computer Science*, vol. 76, pp. 330–335, 2015, doi: 10.1016/j.procs.2015.12.303.